

Driver2Map-pretrain

This is the pretraining code for "Pretrained Prior for Map Refinement" of [Driver2Map](#)

Adapted from [P-MapNet](#)

Environment

1. Create conda environment:

```
conda env create -f environment.yml
conda activate pmapnet
```

2. Install pytorch:

```
pip install torch==1.9.0+cu111 torchvision==0.10.0+cu111 torchaudio==0.9.0 -f
https://download.pytorch.org/whl/torch_stable.html
```

3. Install dependencies

```
pip install -r requirements.txt
```

Dataset Preparation

Download [nuScenes](#) and put it to `./dataset/` folder.

Final Folder Structure

```
Driver2Map-pretrain
|-- config/
|-- data_osm/
|-- model/
|-- random_masks/
|-- tools/
|-- dataset/
|   |-- maps/
|   |-- samples/
|   |-- sweeps/
|   |-- v1.0-trainval/
```

Pretrain the "Pretrained Prior for Map Refinement" module

1. Get the base parameter file, which is trained on ImageNet:

[pretrain-base.pt](#)

2. Edit `./config/hd_pretrain_60m.py`. Make changes to `dataroot`, `version`, etc. Especially change the `vit_base` to the path of `pretrain-base.pt` you had just downloaded.
3. Run:

```
CUDA_VISIBLE_DEVICES=0 python train_HDPrior_pretrain.py --config  
./config/hd_pretrain_60m.py
```